

# Applied NLP

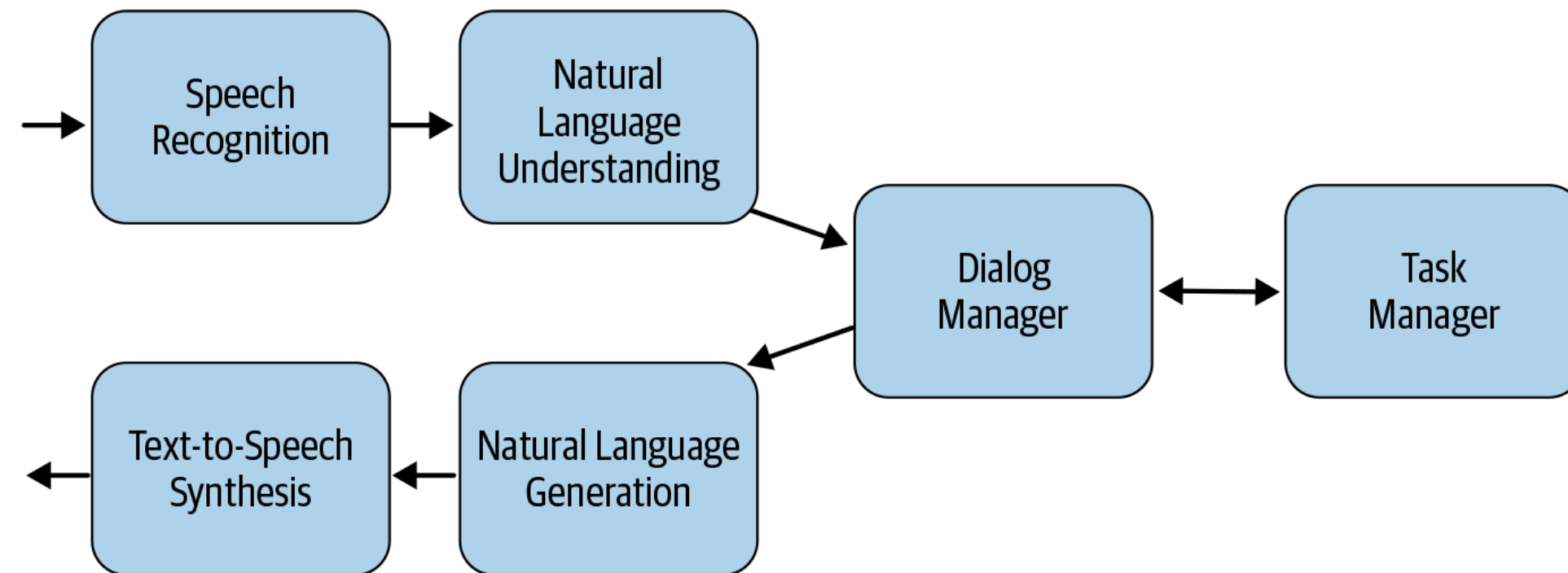
DATA 641

## Lecture 12 — Basic Components of a Chatbot

\*Practical Natural Language Processing: A Comprehensive Guide to Building Real-World NLP Systems 1st Edition, Sowmya Vajjala, Bodhisattwa Majumder, Anuj Gupta, Harshit Surana, O' Reilly and Natural Language Processing in Action, Understanding, analyzing, and generating text with Python, Hobson Lane, Cole Howard, Hannes Hapke, First edition, MANNING

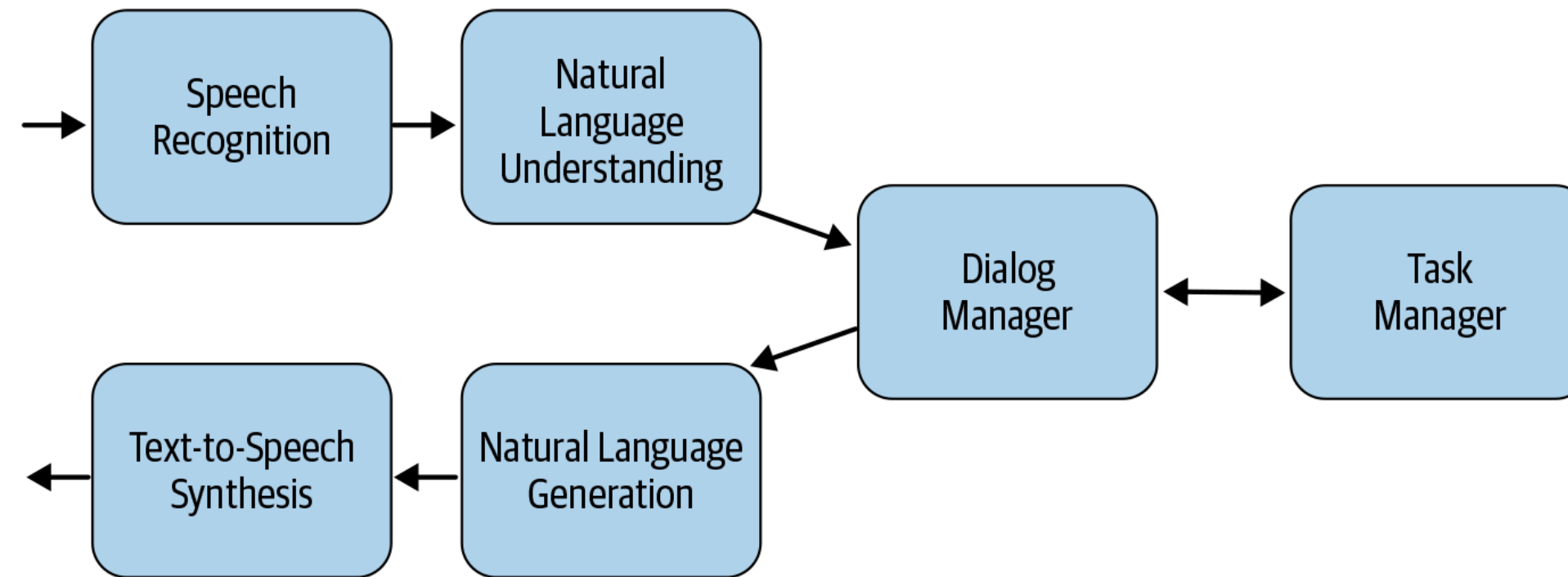
# A Pipeline for Building Dialog Systems

We discussed various NLP tasks, such as **classification** and **entity detection**. Now, we will utilize some of them to describe an example pipeline to build a dialog system.



- **Speech recognition:** Usually, the dialog system works as an interface between human and machine, so the input into the dialog system is human speech. Speech recognition algorithms transcribe speech to natural text. In industrial dialog systems, state-of-the-art speech-to-text models are used.
- **Natural language understanding (NLU):** After transcribing, the system tries to analyze and "understand" the transcribed text. This module encompasses various natural language understanding tasks. Examples of such tasks are sentiment detection, named entity extraction, coreference resolution, etc.

# A Pipeline for Building Dialog Systems



- **Dialog and task manager:** Once we obtain information from the input, a dialog manager, gathers and systematically decides which pieces of information are important or not. A dialog manager is a module that controls and guides the flow of the conversation.
- **Natural language generation:** Finally, as the dialog manager decides a strategy for responding, the natural language generation module generates a response in a human-readable form according to the strategy devised by the dialog manager. The response generator could be template based or a generative model learned from data.

# Dialog Systems in Detail

Below is the terminology used in dialog systems and chatbot development more broadly.

- **Dialog act or intent:** This is the aim of a user command. In traditional systems, the intent is a primary descriptor. Often, several other things, such as sentiment, can be linked to the intent. The intent is also called a "dialog act" in some literature.
- **Slot or entity:** This is the fixed ontological construct that holds information regarding specific entities related to the intent. The information related to each slot that is surfaced in the original utterance is "value." The slots and value together are sometimes denoted as an "entity."
- **Dialog state or context:** A dialog state is an ontological construct that contains both the information about the dialog act as well as state-value pairs. Similarly, context can be viewed as a set of dialog states that also captures previous dialog states as history.

Intent: orderPizza

“get me a medium pizza with extra cheese”

Entity Entity

Intent: getStockQuote

“how is dow jones doing today?”

Entity Entity